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Visual Traversability Analysis for Micro Planetary Rover

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***Abstract*—Traversability analysis is a significant and challenging problem for planetary rover's navigation and safeguard on rocky and bumpy terrain. Moreover, the growing requirement for the micro rover in future planetary missions demands robust traversability analysis upon limited sensors and field of perception. In this paper, a framework for stereo based terrain mapping and traversability analysis in autonomous navigation is developed for the micro planetary rover. In this work, a Joint Reliability based Disparity Expansion (JRDE) algorithm is proposed to obtain accurate 3D information of the local terrain. Focusing on the low complexity and robust performance, we report a hierachical Digital Elevation Map (DEM) based method to achieve the traversability map with relative index value, which consists of elevation thresholding and estimation of slope and roughness. Path planning is subsequently fulfilled upon the traversability map. This method is validated with a set of planet-surface like scenes, while complete onboard system has also been proved fast and effective in artificial rocky environment.**

I. Introduction

AUTONOMOUS navigation is a key problem in the research of planetary rover, which encounters great challenge of the bumpy and rocky planetary terrain. When traversing task regions on planetary surface, the rover should be capable of perceiving surrounding terrains and avoiding hazards, e.g. colliding with rock obstacles, falling into ditch and travelling on steep slope, which might cause slippage or even overturn the rover. However, the nature of planetary surface, such as highly unstructured terrain, rough and bumpy ground, makes the problem hard, meanwhile, other factors, like illumination and shadow, also make the problem worse. Therefore, a robust traversability analysis algorithm based on accurate terrain reconstruction is indispensable for autonomous navigation.

Nowadays, considerable research has been conducted on this problem, in which both lidar and stereo vision are widely used. Although lidar can provide longer range, higher rate and more precise terrain information, the high power consumption makes it unavailable to planetary rovers. Stereo vision system well fits for the planetary rover application, for it is mainly comprised of passive cameras, which consume low volume and energy. However, stereo vision usually suffers poor illumination and costly computation. Hence, a fast stereo algorithm, which is robust to illumination, noise and occlusion, is the basis for visual traversability analysis.

The first practical micro planetary rover - the small sized Sojourner (0.63m×0.48m×0.28m) [1] - uses stereo cameras and infrared laser stripers to detect hazards, but only traveled about 30 meters totally on Mars in 1997. Following rovers, like NASA Spirit and Opportunity Mars Exploration Rovers (MER) with larger size(1.6m×2.3m×1.5m) [2], usually own better traversing capacity, greater autonomy, but they also require much more payload which also encumbers further missions to some degree. Therefore, the micro, lightweight but capable rover should be preferable in future planetary missions, since it can save payload or be deployed in teams to further planets at a time, one typical design in this purpose is JAXA's Micros rovers [3]. This paper focuses on the feasible method of visual traversability analysis for the micro rover, in order to guarantee the rover's reliable autonomy for safe and efficient exploration. The solution for the micro planetary rover should consider following specific conditions:

1) The autonomy relies on input of limited sensors, e.g. IMU for localization and only one stereo camera pair for both hazard detection and navigation, different from that multiple stereo cameras are available on MER.
2) Visual perception is restricted to a narrow field of view due to the small rover size and low height of camera.
3) Avoidance of potential hazards is prior to finding a shortcut, on account of its limited traversing capacity.

In a word, to deal with the visual perception and navigation problem, the micro planetary rover can be described as the small size rover (similar to Sojourner) with relatively limited sensors (e.g. stereo camera and IMU), perception range and traversing capacity. To achieve traversability analysis and following planning, all these features require accurate stereo algorithm, reliable terrain modeling and robust hazards detection. When traversing on planetary surface, the micro rover should perceive the new surroundings and update local map before replanning and action. Although global waypoints should be manually designated according to planetary image and mission requirement, the journey between these global waypoints should be autonomously fulfilled by the rover as above, following the Sense-Plan-Act paradigm. When entering into a region, the rover stops and captures images of frontal terrains. Then terrain's 3D information can be obtained via stereo dense matching, and used to establish the DEM to represent the local terrain. Based on the DEM, each local map cell's traversability is assessed through elevation thresholding and synthetical estimation of

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slope and roughness if necessary, then given corresponding traversability index value. And this local traversability map will be integrated into onboard map for path planning. On account of rover's safe traverse along the path, the rover will move only a short range before the next perception and replanning to ensure itself always within perceived area. Above procedure repeats until that the IMU data indicates arrival at the designated waypoint.

Our proposed method focuses on the local traversability analysis, which provides autonomy for traverse from one global waypoint to the next, as shown in Fig.1. This method has also been demonstrated fast and feasible in rocky simulation environment as a complete onboard solution. Contributions of this paper are mainly as follows:

- Joint Reliability based Disparity Expansion (JRDE) algorithm is proposed to guarantee the accuracy of range data.
- Efficient hierachical DEM-based traversability analysis method is developed, incorporating outlier rejection, elevation thresholding, estimation of slope and roughness to ensure the reliability.
- Complete onboard system is developed and proved fast and effective in artificial rocky environment.

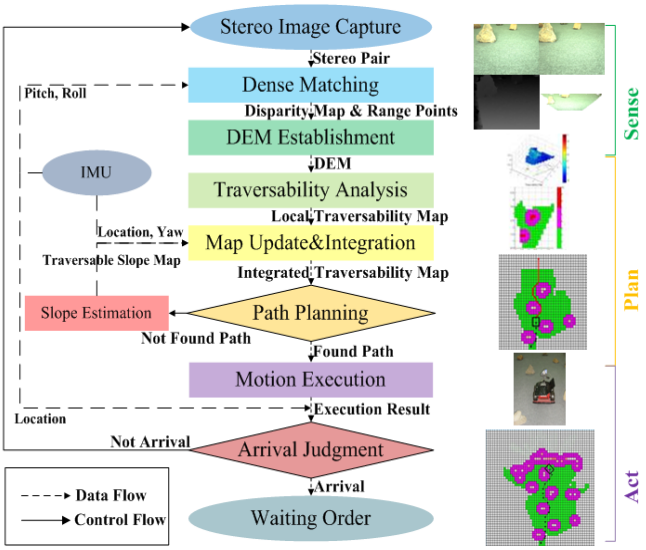


Fig. 1. Procedure of visual traversability analysis method.

II. Related Works

Robust autonomous navigation on rough planetary surface has to depend on reliable terrain analysis model. However, planetary terrains are quite different from ordinary situations. For one thing, the highly rough and unstructured terrains cannot be easily symbolized as indoor scenes, for another, they are short of textures and features from natural or manmade objects on land. To solve this problem, several progresses have been achieved, and two major types among them are most notable. One is based on CMU's Morphin algorithm [4], in which the traversability is calculated along different arcs corresponding to different steering angles. Each arc's traversability is determined by a weighted sum of the map cells' roll, pitch and roughness along the arc, figured as certainty values. One success of this method is NASA's MER program, developing the GESTALT system based on the Morphin algorithm [2]. The other type of methods, proposed by JPL, focuses on using fuzzy logic to construct Rule-Based Traversability Index [5]. The Traversability Index is used for classifying planetary surfaces and provides a simple means for incorporating the terrain quality data. Moreover, there are also other methods, such as the neural network based architecture [6] and a probabilistic modeling technique for high-speed rough-terrain analysis of outdoor mobile robots [7]. Except for only several methods, like the GESTALT, which has been validated on real planetary environment, many other methods still suffer degradation due to the complex, unstructured terrain and unavoidable uncertainty in measurement, although they are mostly experimentally shown accurate in well-known terrain.

However, the platform which GESTALT system actually relies on is quite different from our focused micro rover, as discussed in Chapter 1. To satisfy the requirements of micro rover, we are interested in traversability result for proper path choice rather than particular terrain conditions. Although we also utilize the grid based assessment similar to the GESTALT, our method is actually distinct from GESTALT: we propose JRDE algorithm to improve the accuracy and smoothness of calculated range data. Taking account of the limited perception range and traverse capacity of micro rover, we work out a hierarchical DEM-based traversability assessment by elevation thresholding and optional estimation of slope and roughness, which is lightweight in calculation and feasible to avoid obstacles and steep slopes. Outlier rejection is included to enhance the reliability, and the optional estimation of slope and roughness is able to find out a relatively traversable path when it is trapped in hazardous terrains such as crater. Details will be introduced as follows.

III. Stereo Visual Terrain Perception

Visual perception is to acquire terrains' 3D distribution from the tilt-down stereo cameras, which is challenging in complex situations, such as illumination change, occlusion, noise and shadow. Considering that traditional methods are often vulnerable to above situations, we propose the JRDE method based on the disparity expansion algorithm in [8] to fix the problems. Cameras are calibrated as [9], and stereo rigs are rectified to ensure epipolar lines horizontally aligned. The procedure of this algorithm is as follows:

$$\begin{cases} \Phi(u,v,dp) \le k \times \Phi(u,v,d), \forall d \ne dp \\ \Phi(u,v,dp) \le k \times \Phi(u',v,d'), \forall u'+d' = u+dp \end{cases} \quad (1)$$

Above all, points (u, v, dp) satisfying (1) will be selected as candidate seeds, where dp denotes the disparity, Φ denotes similarity measurement of NCC function, and ratio k is determined by the required seeds number and error rate of the seeds' disparities, the disparity error rate decreases when k is reduced, $k \in (0, 1]$. Then, the candidates are divided to two parts: $S_u=\{(u, v, dp) \mid v < h/2\}$ and $S_d=\{(u, v, dp) \mid v \ge h/2\}$, where h denotes the image height. Maximal surface extraction algorithm in [10] is used to extract the disparity surface D_u and D_d from S_u and S_d. The candidates will be

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accepted as seeds if they satisfy (2) in the square region of (2W+1)×(2W+1):

$$\sum_{i=u-W}^{u+W}\sum_{j=v-W}^{v+W}(D_u(i,j)-D_d(i,j))^2=0 \quad (2)$$

To guarantee the accuracy and reliability, the disparity expansion should follow some constraints. On the one hand, the expansion is ought to be sorted according to seeds' stereo corresponding reliability. The more corresponding between stereo rigs, the higher priority the seeds own in expansion. This correspondence reliability can be evaluated by the NCC coefficient, as (3). On the other hand, seeds expansion along edge might cause error disparities across the edge and damage the terrain reconstruction result. To fix this problem, we eliminate the seeds with high edge response as in SIFT [11], where Hessian Matrix is used to calculate the principle curvatures, and the seeds will be judged as on edge and eliminated if they don't satisfy (4). For the rest seeds, we also use (5) to assess their edge response reliability. The lower the anti-edge reliability is, the closer it is to edge, and the priority in expansion should be decreased.

$$R_{corr}\propto\Phi(u,v,d) \quad (3)$$

$$\frac{Tr(H)^2}{Det(H)}<\frac{(r+1)^2}{r}\ \cap\ Det(H)>0 \quad (4)$$

$$R_{anti-edge}\propto\left(\frac{(r+1)^2}{r}-\frac{Tr(H)^2}{Det(H)}\right)\Big/\frac{(r+1)^2}{r} \quad (5)$$

Therefore, in order to guarantee the accuracy and robustness to illumination, noise and occlusion, we utilize the Joint Reliability of correspondence and anti-edge response to restrict the disparity expansion as (6), where ω is the weight, e.g. ω=0.7. The expansion is sorted following the rank of their Joint Reliability, and expansion of each seed pixel includes its four neighbor pixels: if one neighbor's disparity does not exist, it will be set as (7), and this neighbor pixel will be regarded as seed, its reliability value will be calculated and inserted into the sorted sequence; if the disparity d_0 exists, the new disparity will be obtained as (8). This procedure continues until all seeds are covered. The disparity map is finally fitted by quadratic function to achieve sub-pixel accuracy and processed using adaptive smooth filtering [12].

$$\text{Joint Reliability}=\omega R_{corr}+(1-\omega)R_{anti-edge} \quad (6)$$

$$d'=\arg_{dp}\max(\Phi(u',v',dp)),dp\in\{d-1,d,d+1\} \quad (7)$$

$$d'=\arg_{dp}\max(\Phi(u',v',dp)),dp\in\{d-1,d,d+1,d_0\} \quad (8)$$

Finally, terrain's 3D construction is accomplished. As shown in Fig. 2, this method is competent for unstructured terrain perception, e.g. roughness, rocks, and bumpy ground.

IV. Terrain Mapping and Traversability Analysis

Terrain mapping is to represent local terrain around rover by grid map. Terrain's 3D points are projected onto horizontal world plane and layered within each grid for elevation estimation. Upon this hierarchical DEM, final traversability map is accomplished by elevation thresholding and optional slope estimation, integrating traversability index for path planning. Actually, every time a stereo pair has been obtained, only the terrain within small field of view is maintained and integrated to onboard map, and onboard map only maintains constant size of surrounding traversability information and keeps updating while traversing. When rover runs to the map border, old map will be replaced by a new one, and rover's instant location will be the origin of the new map. All these local maps share a common coordinate frame.

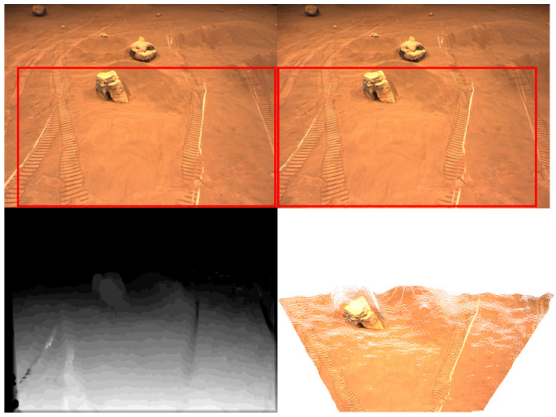


Fig. 2. Result of stereo visual terrain perception. Top: input stereo pair, red rectangle indicates effective region where disparity can be accurately estimated. This region is predefined through experiments. Bottom: disparity map obtained by stereo matching (left) and terrain's 3D point cloud as reconstruction result (right).

A. DEM Building

When a patch of region perceived, the terrain's 3D data in camera coordinate frame is obtained by disparity calculation, then they are translated into world frame according to rover's current location and pose. Subsequently, all 3D points are projected onto horizontal world plane and distributed into each cell. If the particle number distributed in one cell is less than threshold N, which is related to image resolution and grid size, this cell will be labeled unknown for information deficiency. The DEM will be built as follows.

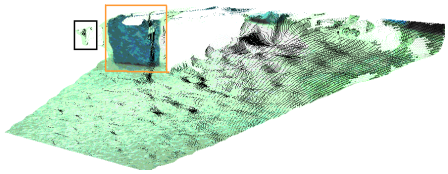


Fig. 3. Outliers in terrain's 3D point cloud: orange rectangle indicates a rocky obstacle, and black rectangle indicates suspending outliers.

Within the 3D point cloud, outliers resulted from error disparities usually emerge suspending in the air, which might cause false elevation in DEM, as shown in Fig. 3. (Usually, these particles will not appear individually for their disparities were filtered in disparity estimation.) To remove these suspending particles, we perform the hierarchical DEM calculation: divide one cell's particles vertically into *n* layers from the bottom points to top ones, e.g. *n*=5. These layers are searched from top to down, if one layer contains much more particles than its neighboring lower layer as (9), particles in this layer and above will be judged suspending and removed,

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and the underlying layers won't be involved into DEM until (10) is satisfied, where $N_{\text{i layer}}$ indicates particle sum in *the i*th layer, and $\tau_1>1.0$, $\tau_2<1.0$, e.g. $\tau_1=1.8$, $\tau_2=0.5$.

$$N_{i\ \text{layer}} \Big/ N_{i-1\ \text{layer}} > \tau_1 \tag{9}$$

$$N_{i+1\ \text{layer}} \Big/ N_{i\ \text{layer}} < \tau_2 \tag{10}$$

Elevation of each cell (h_{cell}) is estimated by the average elevation of particles in remaining top layer ($\overline{h}_{\text{top layer}}$), as (11), where $N_{\text{top layer}}$ denotes the particle sum in remaining top layer, Y_{i} is the elevation of each particle.

$$h_{\text{cell}} = \overline{h}_{\text{top layer}} = \sum_{i=1}^{N_{\text{top layer}}} Y_i \Big/ N_{\text{top layer}} \tag{11}$$

B. Traversability Estimation

Based on DEM, each cell's traversability will be estimated by elevation thresholding, and classified into four categories: traversable, uncertain, untraversable, unknown. Above all, cells with small elevation change from rover's current position are regarded as "traversable", as (12), where $H_{\text{pos_th}}$ and $H_{\text{neg_th}}$ respectively denotes the positive and negative height threshold that rover can span.

$$H_{\text{pos_th}} \geq h_{\text{cell}} - h_{\text{rover}} \geq H_{\text{neg_th}} \tag{12}$$

If the deviation exceeds the range in (12), this cell would be treated as "untraversable", such as obstacles of rocks or pits. Moreover, rover-sized blobs centered at each "untraversable" cell are also treated as "untraversable". Since rover's location on map is denoted by its center, it will encounter hazards when its center goes into this region. But such blob's outer tier should be regarded as "uncertain", because the blob radius is a bit longer than rover radius and these cells need to be reconsidered for traversing when only rover-sized channel between obstacles is available, although there are still hazards in these regions. Cells outside the field of view or occluded by foreground are "unknown".

C. Integration and Update

$$h = \begin{cases} \lambda \times h_{new} + (1-\lambda) \times h_{old}, if\ \ \mathrm{T}_{old} = Traversable, \\ \qquad\qquad\qquad\qquad \mathrm{T}_{new} = Untraversable \\ \eta \times h_{new} + (1-\eta) \times h_{old}, if\ \ \mathrm{T}_{old} = Untraversable, \\ \qquad\qquad\qquad\qquad \mathrm{T}_{new} = Traversable \\ \sigma \times h_{new} + (1-\sigma) \times h_{old}, otherwise \end{cases} \tag{13}$$

As rover rambles, obtained surrounding traversability should be integrated into the onboard map according to rover's localization and pose, and the overlaps should be updated by new data, for nearer range brings more accurate disparity result. The update strategy is as (13), where T denotes traversability, and η, σ, λ stands for different weights for update. Cell's elevation will be updated by weighted average of the old and new elevation, different weights are chosen on condition, and follows: $0<\eta<\sigma<\lambda<1$.

D. Slope Estimation

In our planning strategy, traversing on relatively flat ground is preferable to climbing slopes, which can avoid potential hazards, such as slippage, falling off or even overturning. Moreover, this strategy also prevents huge dead reckoning error caused by slippage. However, this strategy does not work in some occasion, e.g. when rover slumps into a crater. It might be impossible to find a way out unless the slopes previously labeled as "untraversable" are reconsidered. In such a situation, slope estimation will be performed by plane fitting. Centered at each "untraversable" cell, the rover sized patch is used to perform a least-squares plane fitting according to cells' elevation. If the calculation converges within threshold, the patch will be treated as a slope plane, and its gradient and roughness will be calculated. Here, roughness is denoted by the variance of plane fitting.

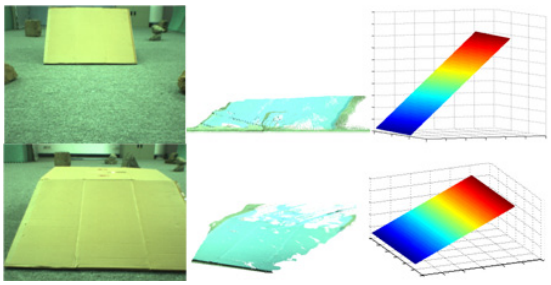


Fig. 4. Results of slope estimation. Left: images of testing slopes. Middle: results of stereo reconstruction. Right: plane fitting results.

TABLE I
CALCULATION RESULTS OF SLOPE ESTIMATION

Scene	Measured Degree	Estimated Degree	Roughness(mm^2)
Slope 1	43.7233	38.2468	367.5160
Slope 2	20.6673	20.2009	41.3120

Fig. 4 and TABLE I show the slope estimation results. Two testing slopes were all labeled as untraversable blobs in elevation thresholding. After plane fitting, if the gradient and roughness is under predefined threshold, the fitting patch would be remarked as "traversable", or else it is still "untraversable". As in TABLE I, slope 1 is "untraversable" because of its gradient (e.g. traversable gradient is within ±30 degrees.), and the roughness (variance) seems large, which should be attributed to some disparity errors caused by too low textures. Slope 2 will be remarked as traversable and taken into path replanning for its low gradient and roughness.

V. PATH PLANNING

In path planning, each cell's traversability is converted to index value for searching algorithm, as shown in Fig. 5. The index value of "unknown" is lower than "uncertain", which will induce rover to explore unknown region when there is no available path in known terrain. Each traversability type is evaluated by a constant positive integral to reduce complexity of searching calculation. Having obtained all index values for building the onboard map, D* Lite algorithm [13] is used to search for an optimal path from current location to destination. D* Lite has higher efficiency and speed than traditional A* and D*, and is competent for fast, incremental planning in our solution. Furthermore, the traversability map of index value is actually the final onboard map. To save

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onboard resources, the DEM calculation and traversability analysis are combined, directly outputting the final index map, and map update and integration will perform on this map. As for the onboard map size, the destination might exceed current map. However, this makes no difference in path planning, since they share common coordinate frame.

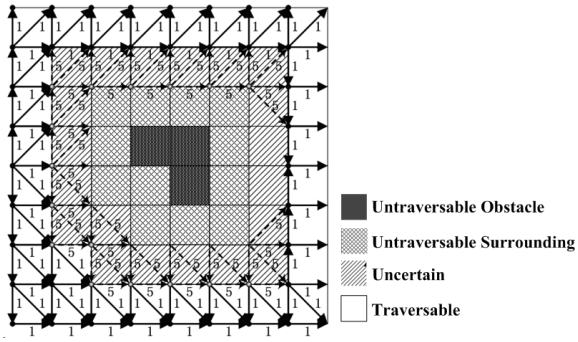


Fig. 5. Index values for different traversabilities: largest for "untraversable" (e.g. 10), medium for "uncertain" (e.g. 5), and minimum for "traversable" (e.g. 1).

VI. Experiment Result

A. Tests on Planet-Surface like Scenes

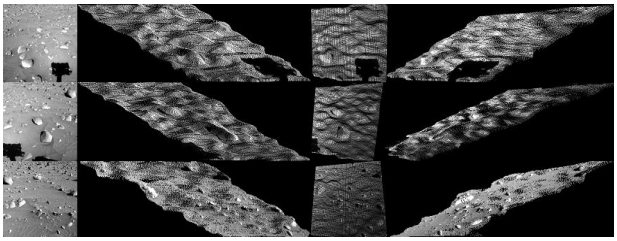


Fig. 6. Results of Mars surface reconstruction, in each row there is the reference image (left) and 3D clouds of reconstructed terrain observed from different viewpoints (left, front and right).

Difficulties within the terrain reconstruction mainly include the high roughness and complexity of terrain, illumination change, low texture and shadow. We use the real Mars surface images from MER [14] to test our algorithm. Although the terrain's slope gradient and real scale cannot be calculated for short of stereo parameters, we can still obtain the reconstruction result in proportion to real terrain. As shown in Fig. 6, our JRDE method is competent for fixing the roughness and small shadow, and achieving reconstruction result vividly reflecting the distribution of small roughness and pebbles. As shown in Fig. 7 (see next page), the robustness to different illumination and low texture of the JRDE method is also proved, and the obtained DEM well accords with the real terrains. Based on the DEM, the traversability map also indicates all untraversable regions.

B. Onboard Test

To test the real performance of our traversability analysis solution, we develop onboard system, involving teleoperation for destination designation, stereo perception, terrain and traversability mapping, path planning, action execution and localization. The system relies on Mobile Robot Pioneer 3-AT, utilizes Digiclops™ Stereo Vision System for perception and a notepad with Intel Core™2 Duo T7300 processor for calculation. Localization mainly depends on the rover's internal IMU.

Test is conducted in the artificial environment with randomly distributed rocks shown in Fig. 8 (see next page). Heading for destination, rover autonomously performs visual traversability analysis and follows planned path to avoid all perceived obstacles. The step between two successive performances of traversability analysis and replanning is set by the range of n cells (e.g. n =2). As shown in Fig. 8, the rover succeeds in detecting all observed obstacles, accurately mapping all untraversable regions, planning optimal path to steer clear of all surrounding obstacles, and arriving at the destination. The traversability mapping result of all obstacles accurately accords with real locations of all rocks, as shown in Fig. 8 and TABLE Ⅱ. The remaining mapping error is mainly attributed to the grid map's precision. Because the map is based on $0.2\times0.2\text{m}^2$ grid, most errors are just at this scale, and these errors can be compensated in the expansion of untraversable blob.

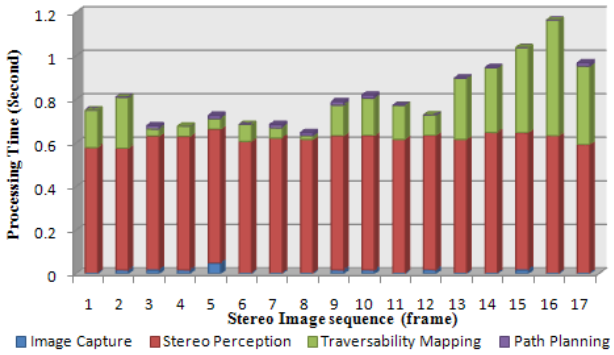


Fig. 9. Processing time of each module.

Processing time of each module in the onboard test is shown in Fig. 9. On the average, visual perception (mainly the JRDE algorithm) cost 613ms, terrain and traversability mapping consumes 182ms, image capture costs 9.2ms, path planning uses 6.4ms. We can see most time is spent in the perception module, ranging from 560ms to 648ms, this is mainly because we use the 320 × 240 pixels image to guarantee the abundant information for terrain mapping. Average time consumption of whole visual traversability analysis including image capture and path planning is 810ms.

VII. Conclusion

In this paper, we propose a visual traversability analysis solution for micro planetary rover. The proposed JRDE algorithm is robust to illumination, small shadow area and capable of obtaining accurate range data of highly rough and unstructured terrain. The hierarchical DEM-based traversability analysis incorporating elevation thresholding and slope estimation have also been proved fast and effective to support robust autonomous navigation and safeguard. The experiments upon both testing scenes and onboard system demonstrate the feasibility and robustness of our method in rocky terrains. Future work will involve acceleration of the JRDE algorithm, further integration of slope estimation and parallel processing architecture.

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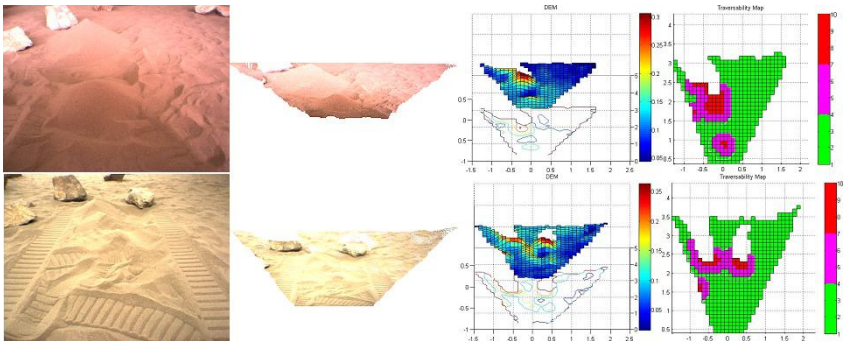


Fig. 7. Tests in planet-surface like scenes: there is reference image, terrain reconstruction result, DEM and traversability map in each row. Each map grid is 20cm×20cm, and color brightness increases with elevation in DEM. In traversability map, purple indicates "untraversable" region, whose outer tier is "uncertain". Untraversable terrains over threshold (e.g. ±20cm) are in red. Green indicates "traversable" and white indicates "unknown".

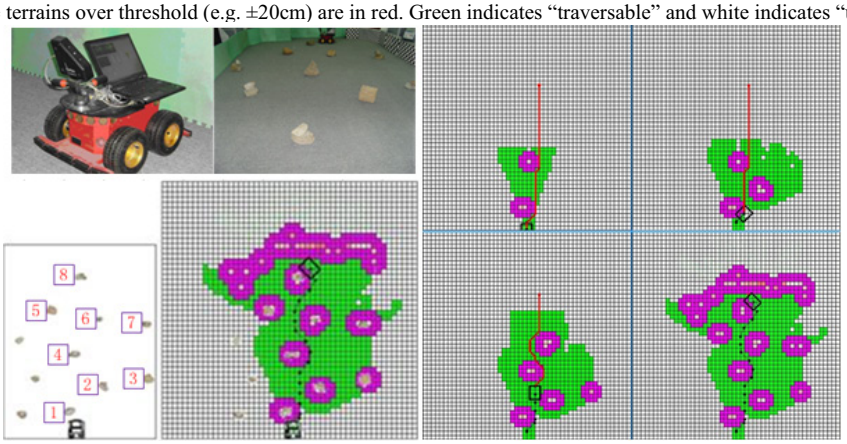


Fig. 8. Onboard test: test bed and artificial rocky environment (top left), comparison of all obstacles' reference locations and mapping results (bottom left), updates of traversability map in onboard test (right). The meaning and parameters of the map are same as Fig. 7. The precise information of each labeled obstacles are shown in Table Ⅱ.

TABLE II
COMPARISON OF OBSTACLE'S REFERENCE LOCATION AND MAPPING RESULT

Obstacle	Reference Location(m)		Mapping Location(m)		Error(m)	
	Central Location	Radius	Central Location	Radius	Center Location	Radius
1	(-0.154, 1.126)	0.192	(-0.200, 1.100)	0.224	0.053	0.032
2	(0.932, 2.243)	0.236	(0.800, 2.100)	0.224	0.195	0.012
3	(2.911, 2.525)	0.209	(3.100, 2.500)	0.141	0.191	0.068
4	(0.018, 3.174)	0.215	(0.200, 3.300)	0.224	0.221	0.009
5	(-0.853, 5.312)	0.237	(-0.800, 5.100)	0.224	0.219	0.013
6	(0.978, 4.975)	0.201	(1.000, 4.800)	0.283	0.176	0.082
7	(2.745, 4.838)	0.255	(2.900, 4.700)	0.316	0.208	0.061
8	(0.137, 6.212)	0.189	(0.300, 6.200)	0.224	0.163	0.035

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